

Q1. (a) CYK Parsing

Given Grammar:

1. $S \rightarrow NP VP$
2. $VP \rightarrow V NP$
3. $NP \rightarrow Det N \mid N$
4. $Det \rightarrow \text{"the"} \mid \text{"a"}$
5. $N \rightarrow \text{"dog"} \mid \text{"cat"} \mid \text{"ball"}$
6. $V \rightarrow \text{"chased"}$

Sentence to Parse: "the dog chased a cat"

- (i) Explain what characteristics of a grammar make it suitable for the CYK algorithm.
- (ii) Apply the **CYK algorithm** to parse the given sentence. Fill all levels of the CYK table step by step, showing how each non-terminal is derived for every cell. Identify the final cell entry and interpret whether the sentence is accepted by the grammar.
- (iii) In the final cell of the table, what non-terminals appear? What does this indicate about the grammatical validity of the sentence?
- (iv) If the grammar is changed to include the rule $VP \rightarrow V$, explain briefly how this change might affect the parsing of the same sentence and mention one new sentence that would now be valid under this modified grammar.

Q1. (b) CYK Parsing

Given Grammar:

1. $S \rightarrow NP VP$
2. $VP \rightarrow V NP$
3. $NP \rightarrow Det N$
4. $Det \rightarrow \text{"the"} \mid \text{"an"}$
5. $N \rightarrow \text{"teacher"} \mid \text{"student"} \mid \text{"apple"}$
6. $V \rightarrow \text{"gave"}$

Sentence to Parse: “the teacher gave an apple”

- (i) Apply the **CYK algorithm** to parse the given sentence. Fill all levels of the CYK table step by step, showing how each non-terminal is derived for every cell. Identify the final cell entry and interpret whether the sentence is accepted by the grammar.
- (ii) In the final cell of the table, what non-terminals appear? What does this indicate about the grammatical validity of the sentence?
- (iii) If the grammar is changed to include the rule $VP \rightarrow V$ (in addition to $VP \rightarrow V NP$), explain briefly how this change might affect the parsing of the same sentence and mention one new sentence that would now be valid under this modified grammar.

Q1. (c) CYK Parsing

Given Grammar:

- 1. $S \rightarrow NP VP$
- 2. $VP \rightarrow V NP$
- 3. $NP \rightarrow Det N \mid N$
- 4. $Det \rightarrow \text{“the”} \mid \text{“a”}$
- 5. $N \rightarrow \text{“dog”} \mid \text{“cat”} \mid \text{“ball”}$
- 6. $V \rightarrow \text{“chased”}$

Sentence to Parse: “the dog chased a cat”

Answer the following questions:

- (i) Apply the **CYK algorithm** to parse the given sentence. Fill all levels of the CYK table step by step, showing how each non-terminal is derived for every cell. Identify the final cell entry and interpret whether the sentence is accepted by the grammar.
- (ii) In the final cell of the table, what non-terminals appear? What does this indicate about the grammatical validity of the sentence?
- (iii) If the grammar is changed to include the rule $VP \rightarrow V$, explain briefly how this change might affect the parsing of the same sentence and mention one new sentence that would now be valid under this modified grammar.

Q2. (a) Semantic Disambiguation in Context-Aware Systems

Read the following scenario carefully and answer:

An autonomous driving assistant receives the sentence:

“The pedestrian is crossing near the bank.”

The system must decide whether *bank* refers to a **riverbank** or a **financial institution**, as the meaning determines the type of warning or response to trigger.

- (i) Explain how the **Lesk algorithm** (a knowledge-based approach using **WordNet**) would resolve this ambiguity by computing overlaps between glosses and contextual words.
- (ii) Describe how a **neural contextual model such as BERT** and an **unsupervised graph-based method like HyperLex** can identify word senses by leveraging context embeddings and co-occurrence networks.
- (iii) Compare the **Lesk (symbolic)**, **HyperLex (unsupervised)**, and **BERT (neural)** approaches in terms of interpretability, scalability, and suitability for real-time disambiguation in low-resource or high-noise settings.

Q2. (b) Lexical Semantics and Polysemy Handling in AI Assistants

A multilingual voice assistant receives the sentence:

“Can you book a flight for me tomorrow?”

The system must distinguish between the *verb* “book” (to reserve) and the *noun* “book” (a physical object) to ensure correct intent classification.

- (i) Explain how a **supervised classifier** could be trained for word-sense disambiguation using contextual features such as neighboring words, POS tags, and dependency relations.
- (ii) Suggest how a **semi-supervised Yarowsky algorithm** could expand the labeled dataset for low-resource languages.
- (iii) Discuss one major **challenge of polysemy** in multilingual NLP and propose a solution that leverages **cross-lingual embeddings**.

Q3.(a) Anaphora Resolution in Narrative Text

Read the following text carefully:

“Meera called Riya after she finished her project. She thanked her for the help and invited her to the celebration.”

- (i) Identify all noun phrases (NPs) and the pronouns that require resolution.
- (ii) Apply **Hobbs’ Algorithm** step by step to determine the antecedents of “she” and “her” in both sentences. Clearly show each traversal and selection step.
- (iii) Using the **Centering Algorithm**, list the **Forward-Looking Centers (Cf)** and **Backward-Looking Centers (Cb)** for each sentence, and derive the preferred center transitions.
- (iv) Compare the results of both algorithms and explain which produces more contextually coherent antecedent mapping for this dialogue.

Q3.(b) Coreference and Discourse Coherence in Event Description

Consider the following passage:

“Amit gave Rahul his notes before the exam. He said they would help him understand the topic better.”

- (i) Identify all noun phrases (NPs) and pronouns that need resolution.
- (ii) Apply **Hobbs’ Algorithm** to determine the antecedents of “he,” “they,” and “him.” Show reasoning for each traversal path.
- (iii) Use the **Centering Algorithm** to find the **Cf** and **Cb** lists for both sentences and analyze the center transitions (Continue, Retain, or Shift).
- (iv) Compare how **Hobbs’** and **Centering** algorithms interpret the discourse structure differently, and discuss which provides a more natural interpretation in human-like conversation.

Q4.(a) Sentiment Analysis for Regional E-Governance Feedback

A state government is developing an **AI feedback analysis system** to process citizen comments in regional languages such as *Marathi, Gujarati, and Kannada*. The system

should automatically classify feedback as positive, negative, or neutral to guide administrative decisions.

- (i) Design an **NLP pipeline** for multilingual sentiment analysis including modules for **text normalization, language identification, feature extraction, and classification**.
- (ii) Explain how **contextual embeddings** (e.g., multilingual BERT or XLM-R) improve sentiment prediction for code-mixed or transliterated data.
- (iii) Discuss one challenge specific to low-resource Indian languages and suggest a method for handling domain-specific vocabulary or sarcasm.

Q4.(b) Abstractive Text Summarization for Health Awareness Articles

A public health organization wants to develop an **automatic summarization system** that generates short, readable summaries of long health-awareness articles in *Hindi and English*. The goal is to make the content easier for rural readers to understand.

- (i) Outline the **components of an abstractive summarization model** — including preprocessing, encoder–decoder framework, attention mechanism, and post-processing.
- (ii) Explain how **semantic coherence and factual consistency** are maintained in abstractive summaries using transformer architectures such as **T5** or **BART**.
- (iii) Identify one major **challenge** when generating summaries in bilingual or low-resource settings and suggest a **data augmentation or transfer learning** approach to overcome it.

Q4.(c) Multilingual Speech-to-Text Summarization System

A news agency wants to develop a **speech-to-text summarization system** for multilingual press conferences in *Hindi, Marathi, and English*.

Design a workflow for such a system, explaining the stages of **speech recognition, language identification, text segmentation, summarization, and evaluation**. Also, discuss one major challenge in maintaining **semantic coherence** and **speaker context** across multiple languages.

(ii) Emotion Detection in Customer Support Chats

A fintech company wants to improve its **AI-based customer support system** by detecting emotional tone in chat messages.

Propose an NLP pipeline that performs **text preprocessing, emotion classification, and context-based response generation**. Discuss one challenge in detecting **subtle emotions** (such as frustration or sarcasm) in short or code-mixed messages.

Q4.(d) Automatic Headline Generation for News Articles

A media startup wants to create an **automatic headline generation tool** for Hindi and English news stories using deep learning models.

Outline a workflow including modules for **text preprocessing, encoder–decoder modeling, attention mechanism, and evaluation** (e.g., ROUGE, BLEU). Explain one major challenge related to preserving **factual accuracy** and avoiding **hallucinated content** in generated headlines.

(ii) Named Entity Recognition for Indian Languages

A government portal wants to build a **Named Entity Recognition (NER)** system to automatically extract names of people, organizations, and locations from multilingual documents.

Design a pipeline covering **data preprocessing, feature representation** (e.g., embeddings), **sequence labeling, and post-processing**. Explain one specific challenge in handling **morphologically rich languages** such as Tamil or Kannada during entity boundary detection.

Q4.(e) Machine Translation

A multinational e-commerce platform needs to provide **real-time, high-quality translations** of product descriptions and customer reviews in **15 different languages**, while preserving **cultural nuances** and **technical jargon**.

Answer the following questions:

- (i) Outline the **architecture (block diagram)** for a **machine translation model** that ensures accuracy across a wide variety of languages, including low-resource languages.
- (ii) Discuss how **advancements like transformer-based models** improve translation quality compared to traditional **rule-based systems**.

- (iii) Explain how the system can handle **domain-specific terms** and **cultural context** without compromising fluency and grammatical correctness.

Q4.(f) Question Answering (QA)

A healthcare organization wants to build a **Question Answering (QA)** system to provide accurate and reliable answers to patient queries about **symptoms, medications, and treatments**, based on a comprehensive **medical knowledge base**.

Answer the following questions:

- (i) Outline the **architecture (block diagram)** for a QA system that ensures accurate extraction and comprehension of **domain-specific medical knowledge**.
- (ii) Explain how **transformer-based models** (e.g., **BERT, T5**) and **retrieval-augmented generation (RAG)** improve QA performance in understanding and generating answers.
- (iii) Describe how the model can handle **ambiguous or incomplete queries** to provide meaningful and contextually appropriate responses.
- (iv) Propose techniques to ensure that the QA system prioritizes **accuracy**, avoids **misinformation**, and maintains **reliability** in critical healthcare scenarios.

Q5.(a) Structural Ambiguity in Prepositional Phrase

Consider the following CFG and the sentence to be parsed:

1. $S \rightarrow NP VP$
2. $NP \rightarrow N \mid Det N \mid NP PP$
3. $VP \rightarrow V NP \mid VP PP$
4. $PP \rightarrow P NP$
5. $Det \rightarrow \text{"the"}$
6. $N \rightarrow \text{"man"} \mid \text{"telescope"} \mid \text{"hill"}$
7. $V \rightarrow \text{"saw"}$
8. $P \rightarrow \text{"on"} \mid \text{"with"}$

Sentence: "The man saw the hill with the telescope."

- (i) Explain why this sentence is syntactically ambiguous and describe the two possible parse tree interpretations.
- (ii) Using a **shift–reduce parser**, outline the sequence of **shift** and **reduce operations** for each interpretation, showing how the ambiguity arises.
- (iii) Analyze how the **order of reductions** affects the final parse structure and the meaning of the sentence.

Q5.(b) Ambiguity in Coordination and Attachment

Consider the following CFG and the sentence to be parsed:

1. $S \rightarrow NP VP$
2. $NP \rightarrow Det N \mid Det N Conj NP \mid NP PP$
3. $VP \rightarrow V NP \mid V NP PP$
4. $PP \rightarrow P NP$
5. $Det \rightarrow \text{“the”}$
6. $N \rightarrow \text{“boy”} \mid \text{“girl”} \mid \text{“telescope”}$
7. $V \rightarrow \text{“saw”}$
8. $P \rightarrow \text{“with”}$
9. $Conj \rightarrow \text{“and”}$

Sentence: “The boy and the girl saw the telescope with the lens.”

- (i) Describe the types of syntactic ambiguities present in the sentence (coordination vs. attachment).
- (ii) Use a **shift–reduce parsing** approach to illustrate at least two distinct parse sequences leading to different interpretations.
- (iii) Explain how **semantic cues** (such as plausible role assignments) can help select the most likely parse in a real NLP system.

Q5.(c) Structural Ambiguity and Shift–Reduce Parsing

Context-Free Grammar (CFG):

1. $S \rightarrow NP\ VP$
2. $NP \rightarrow \text{Pronoun} \mid \text{Det}\ N$
3. $VP \rightarrow V\ NP \mid V\ NP\ PP$
4. $PP \rightarrow P\ NP$
5. $\text{Det} \rightarrow \text{the}$
6. $N \rightarrow \text{dog} \mid \text{stick}$
7. $V \rightarrow \text{chased}$
8. $P \rightarrow \text{with}$

Sentence to Parse: “I chased the dog with the stick.”

Answer the following questions:

- (i) Explain why the given sentence is syntactically ambiguous and describe the two possible interpretations based on its parse structure.
- (ii) Draw the two distinct parse trees that illustrate these interpretations clearly.
- (iii) Using a **shift–reduce parser**, outline the sequence of shift and reduce operations for each interpretation, showing how ambiguity arises.
- (iv) Comment on how the **order of reductions** affects the final parse structure and meaning of the sentence.

Q6.(a) Word Sense Disambiguation using Naïve Bayes (Supervised Learning)

A supervised WSD classifier must decide whether the word “**cell**” means (i) *a small room (e.g., prison cell)* or (ii) *a biological unit (e.g., human cell)*.

The following probabilities are learned from a training corpus:

Feature	P(Feature — room)	P(Feature — biology)
word = “prison”	0.5	0.05
word = “organism”	0.1	0.4
word = “microscope”	0.05	0.3
word = “door”	0.3	0.1

Priors: $P(\text{room}) = 0.6$, $P(\text{biology}) = 0.4$

Test Sentence: “The scientist examined the cell under the microscope.”

- (i) Compute the **posterior probabilities** for both senses using the **Naïve Bayes assumption**.
- (ii) Identify the **most probable sense** of “cell” and interpret it in context.

Q6.(b) Naïve Bayes-based Supervised Word Sense Disambiguation

A supervised classifier must determine whether the word “**bat**” refers to (i) *an animal (flying mammal)* or (ii) *a sports instrument*.

The following probabilities are learned from the training corpus:

Feature	P(Feature — animal)	P(Feature — sport)
word = “fly”	0.6	0.05
word = “cricket”	0.05	0.5
word = “nocturnal”	0.4	0.05
word = “ball”	0.1	0.4

Priors: $P(\text{animal}) = 0.45$, $P(\text{sport}) = 0.55$

Test Sentence: “The player swung the bat to hit the ball.”

- (i) Compute the **posterior probabilities** for both senses using the **Naïve Bayes approach**.
- (ii) State which sense is predicted and briefly explain the reasoning.

Q7.(a) Semantic Relations in WordNet (Conceptual and Analytical)

WordNet is a large lexical database that organizes English words into sets of synonyms (synsets) and defines semantic relations among them.

- (i) Explain how different semantic relations such as **hypernymy**, **meronymy**, **antonymy**, and **troponymy** are structured in WordNet, providing one example for each from real-world usage.
- (ii) Describe how **polysemy** and **homonymy** are handled in WordNet’s synset-based architecture, and how this distinction supports **Word Sense Disambiguation (WSD)**.
- (iii) Discuss how the representation of these relations enables WordNet to support **semantic similarity** and **text entailment** tasks in NLP applications.

Q7.(b) Role of WordNet in Semantic Understanding and Inference

Consider how WordNet contributes to deeper semantic interpretation in NLP tasks.

- (i) Illustrate with suitable examples how WordNet captures **entailment**, **causation**, and **antonymy** between verbs and adjectives. For example: “snore” entails “sleep”, “buy” causes “own”, and “hot” is the antonym of “cold”.
- (ii) Explain how hierarchical relations like **hypernymy–hyponymy** support taxonomy building, and how **meronymy** aids information retrieval or question answering systems.
- (iii) Analyze the limitations of WordNet in representing real-world semantics — such as cultural context, idioms, or multiword expressions — and suggest how extended resources like **WordNet++** or **BabelNet** address these limitations.

Q8.(a) Coreference Resolution with Syntactic and Semantic Constraints

Read the following sentences carefully and answer the questions that follow:

(S1) *Ravi helped Suresh to move his luggage before he left for the airport.* (S2) *The cat chased the dog because it was scared.*

- (i) Identify all **noun phrases (NPs)** and **pronouns** that require coreference resolution in both sentences.
- (ii) Explain how **syntactic constraints** (number, person, case, reflexivity) and **semantic constraints** (verb restrictions, world knowledge) influence the valid antecedents for “he” and “it”.

- (iii) Resolve the pronoun ambiguities using syntactic and semantic reasoning and justify your interpretations.
- (iv) Discuss how **recency** and **grammatical role salience** affect the preferred antecedents in each case.

Q8.(b) Discourse-Level Pronoun Resolution and Pragmatic Inference

Consider the following short discourse:

Priya met Neha after she submitted her project. She thanked her for the help.

- (i) Identify all **noun phrases** and **pronouns** that need resolution.
- (ii) Using syntactic and semantic cues, determine the possible antecedents for each pronoun (“she”, “her”) and explain the resulting ambiguities.
- (iii) Apply **pragmatic reasoning** (world knowledge and discourse salience) to infer the most coherent interpretation.
- (iv) Explain how **parallelism** and **repeated mention** preferences influence pronoun resolution in automatic NLP systems.

Q9.(a) Email Spam Classification using Naïve Bayes

Classify an email as **Spam** or **Not Spam** using the **Naïve Bayes classification algorithm**.

Training Dataset:

Email	Label
Free iPhone, click now!	Spam
Meeting at 3pm in the conference room	Not Spam
Win a prize, click here!	Spam
Let’s confirm our meeting for tomorrow	Not Spam
Discount offer on your next purchase!	Spam

Test Email: “The conference meeting was moved to 4pm.”

- (i) Apply the Naïve Bayes algorithm to classify the test email as *Spam* or *Not Spam*.
- (ii) Compute the posterior probabilities and justify your classification decision.

Q9.(b) Sentiment Analysis using Naïve Bayes

Classify a movie review as **Positive** or **Negative** using the **Naïve Bayes classification algorithm**.

Training Dataset:

Review	Label
An amazing movie with a great plot!	Positive
Terrible storyline and poor acting.	Negative
Fantastic visuals and brilliant soundtrack!	Positive
Boring plot and bad character development.	Negative
A truly breathtaking and memorable experience!	Positive

Test Review: “The plot was engaging, but the visuals were disappointing.”

- (i) Use the Naïve Bayes model to classify the test review as *Positive* or *Negative*.
- (ii) Show intermediate probability calculations and provide reasoning for your final prediction.